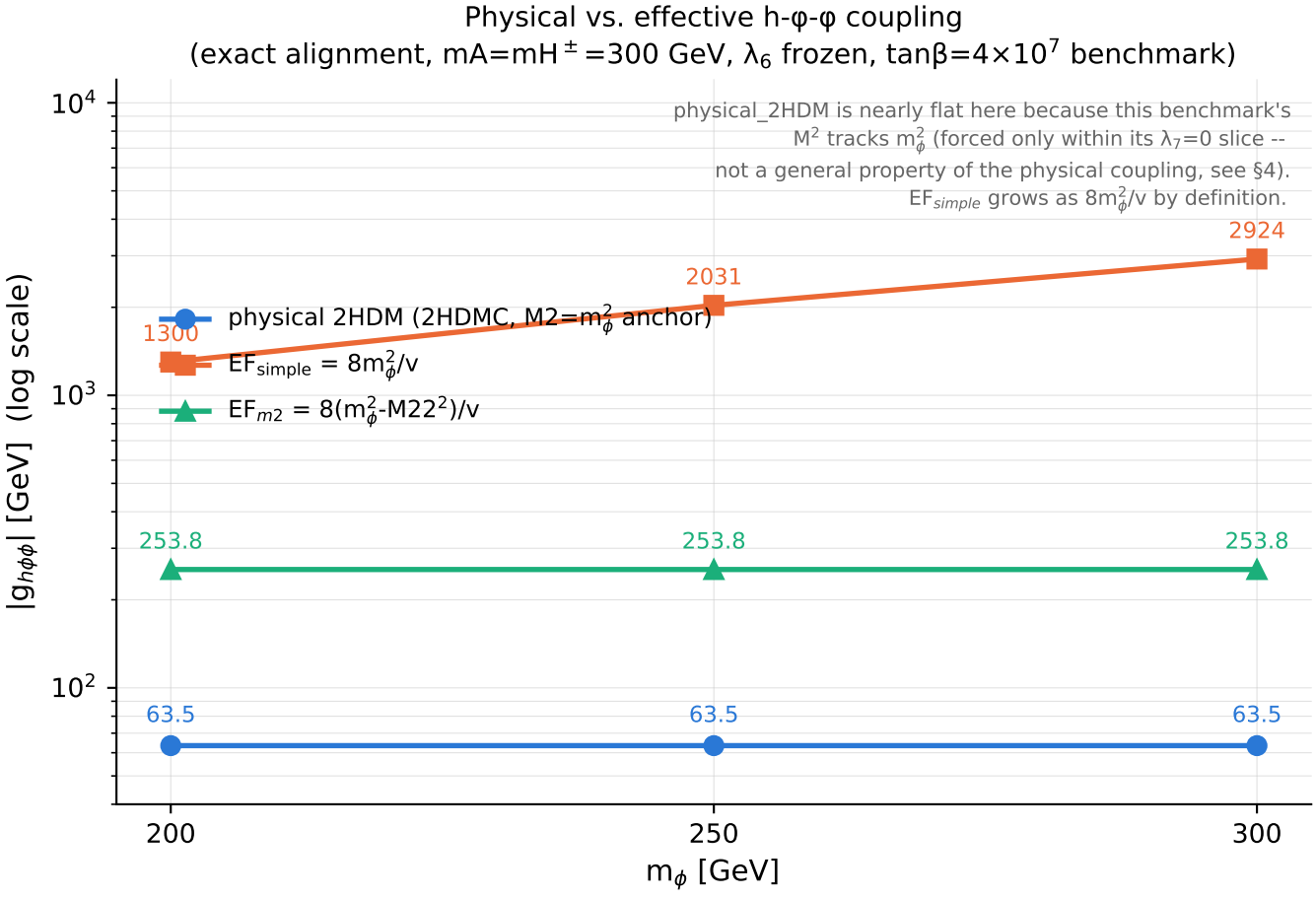
Physical vs. effective h-φ-φ coupling
(exact alignment, mA=mH±=300 GeV, λ6 frozen, tanβ=4×10^7 benchmark)
physical_2HDM is nearly flat here because this benchmark's
M² tracks m²φ (forced only within its λ7=0 slice --
not a general property of the physical coupling, see §4).
EF_simple grows as 8m²φ/v by definition.
physical 2HDM (2HDMC, M2=m²φ anchor)
EF_simple = 8m²φ/v
EF_m2 = 8(m²φ-M22²)/v
2924
2031
1300
253.8
253.8
253.8
63.5
63.5
63.5
10^4
10^3
10^2
200
250
300
|g_hφφ| [GeV] (log scale)
m_φ [GeV]